

Exercises: Table Relations

You can check your solutions in the [Judge system](#).

1. One-To-One Relationship

Create **two** tables and use appropriate data types.

Example

Persons			
PersonID	FirstName	Salary	PassportID
1	Roberto	43300.00	102
2	Tom	56100.00	103
3	Yana	60200.00	101

Passports	
PassportID	PassportNumber
101	N34FG21B
102	K65LO4R7
103	ZE657QP2

Insert the data from the example above. Alter the **Persons** table and make **PersonID** a **primary key**. Create a **foreign key** between **Persons** and **Passports** by using the **PassportID** column.

2. One-To-Many Relationship



Create **two** tables and use appropriate data types.

Example

Models		
ModelID	Name	ManufacturerID
101	X1	1
102	i6	1
103	Model S	2
104	Model X	2
105	Model 3	2
106	Nova	3

Manufacturers		
ManufacturerID	Name	EstablishedOn
1	BMW	07/03/1916
2	Tesla	01/01/2003
3	Lada	01/05/1966

Insert the data from the example above and add **primary keys** and **foreign keys**.

3. Many-To-Many Relationship



Create **three** tables and use appropriate data types.

Example

Students	
StudentID	Name
1	Mila
2	Toni
3	Ron

Exams	
ExamID	Name
101	SpringMVC
102	Neo4j
103	Oracle 11g

StudentsExams	
StudentID	ExamID
1	101
1	102
2	101
3	103
2	102
2	103

Insert the data from the example above and add **primary keys** and **foreign keys**. Keep in mind that the table "StudentsExams" should have a **composite primary key**.

4. Self-Referencing

Create **one** table and use appropriate data types.

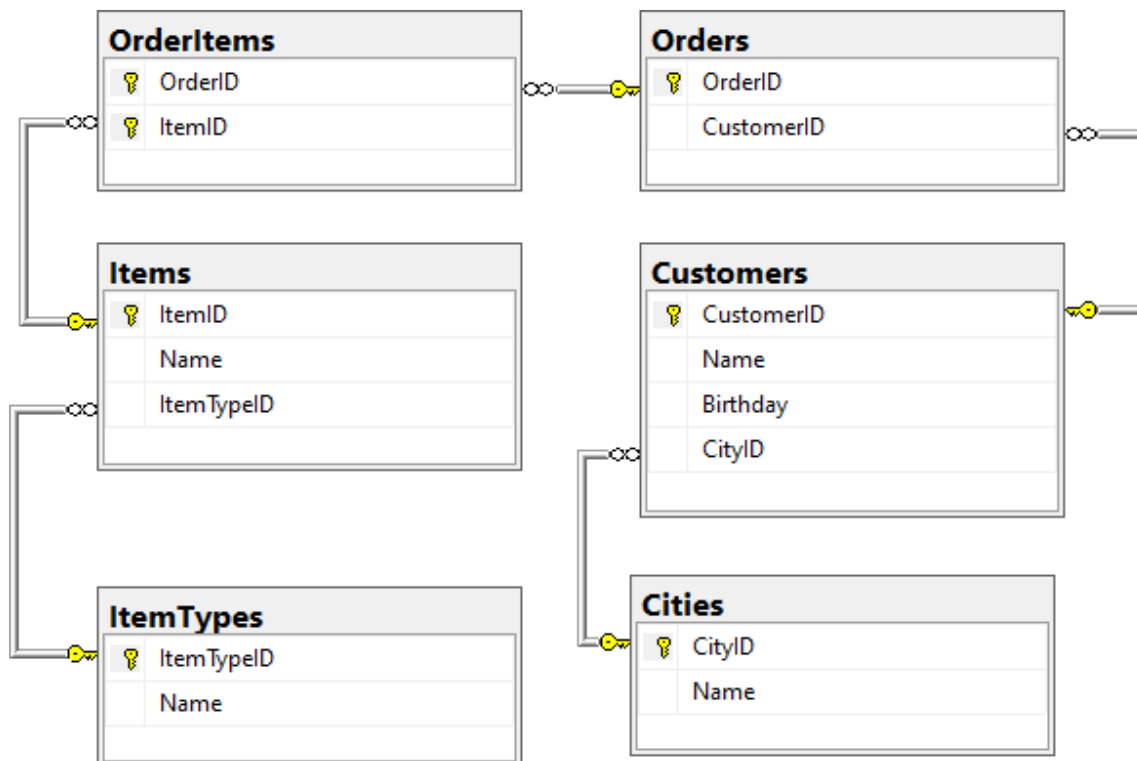
Example

Teachers		
TeacherID	Name	ManagerID
101	John	NULL
102	Maya	106
103	Silvia	106
104	Ted	105
105	Mark	101
106	Greta	101

Insert the data from the example above and add **primary keys** and **foreign keys**. The **foreign key** should be between **ManagerId** and **TeacherId**.

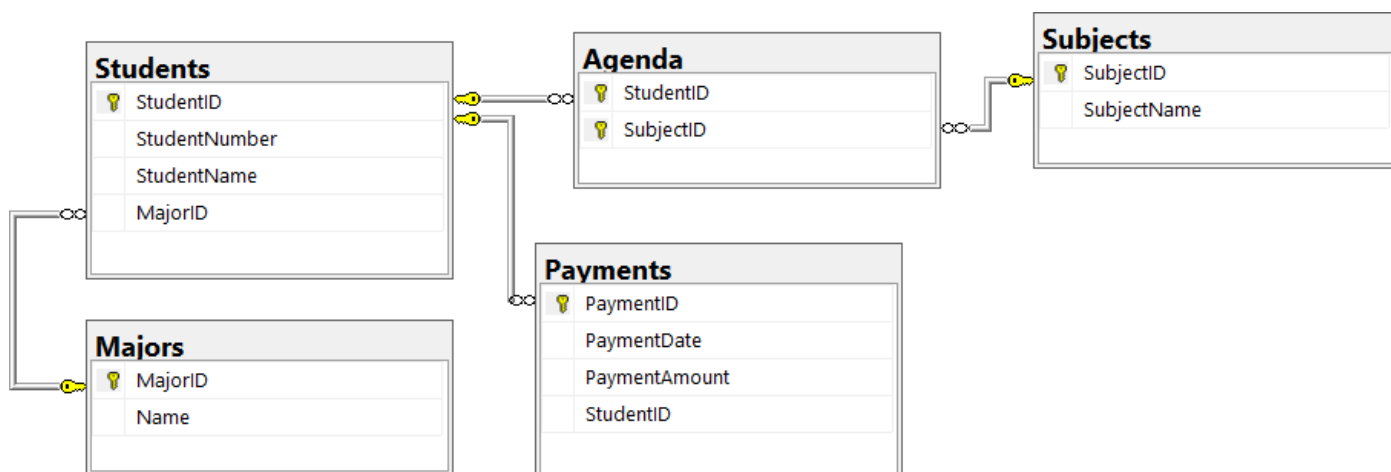
5. Online Store Database

Create a new database and design the following structure:



6. University Database

Create a new database and design the following structure:



7. SoftUni Design

Create an **E/R Diagram** of the SoftUni Database. There are some **special relations** you should **check out**:

- Employees are **self-referenced** (**ManagerID**)
- Departments have **One-to-One** with the Employees (**ManagerID**)
- Employees have **One-to-Many** (**DepartmentID**)

You might find it interesting how it looks on the diagram.

8. Geography Design

Create an **E/R Diagram** of the Geography Database.

9. *Peaks in Rila



Display **all peaks** for "**Rila**" mountain. Include:

- **MountainRange**
- **PeakName**
- **Elevation**

Peaks should be **sorted** by **elevation descending**.

Example

MountainRange	PeakName	Elevation
Rila	Musala	2925
...	...	...